

Optimization of Graph Squares - Proofs

Eddie Xu

`eddie.xu@student.unimelb.com.au`

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1 Formulations

Let known constants $g_{ij} \in \{0, 1\}$ come from the adjacency matrix of the square A_G and let $M \in \mathbb{Z}_{>0}$ be a sufficiently large constant. The variables in the program are x_{ij} , y_{ikj} , z_{ij} where x_{ij} form the ij -th entry of the adjacency matrix of the root A_H .

1.1 Integer Program

Variables:

$$x_{ij} \in \mathbb{Z}_{\geq 0}, \quad \forall i, j \in \{1, \dots, n\}, \quad (1)$$

$$y_{ikj} \in \mathbb{Z}_{\geq 0}, \quad \forall i, k, j \in \{1, \dots, n\}, \quad (2)$$

$$z_{ij} \in \mathbb{Z}_{\geq 0}, \quad \forall i, j \in \{1, \dots, n\}. \quad (3)$$

Constraints:

$$(1) \text{ Indicator argument: } z_{ii} = \sum_{k=1}^n y_{iki} - \sum_{k=1}^n x_{ik}, \forall i \in \{1, \dots, n\}, \quad (4)$$

$$z_{ij} = \sum_{k=1}^n y_{ikj} + x_{ij}, \quad \forall i \neq j \in \{1, \dots, n\}, \quad (5)$$

$$(2) \text{ Indicator function: } z_{ij} \leq M g_{ij}, \quad \forall i, j \in \{1, \dots, n\}, \quad (6)$$

$$z_{ij} \geq g_{ij}, \quad \forall i, j \in \{1, \dots, n\}, \quad (7)$$

$$(3) \text{ Product: } y_{ikj} \leq x_{ik}, \quad \forall i, k, j \in \{1, \dots, n\}, \quad (8)$$

$$y_{ikj} \leq x_{kj}, \quad \forall i, k, j \in \{1, \dots, n\}, \quad (9)$$

$$y_{ikj} \geq x_{ik} + x_{kj} - 1, \quad \forall i, k, j \in \{1, \dots, n\} \quad (10)$$

$$(4) \text{ Upper Bound: } x_{ij} \leq 1, \quad \forall i, j \in \{1, \dots, n\}, \quad (11)$$

$$y_{ikj} \leq 1, \quad \forall i, k, j \in \{1, \dots, n\}, \quad (12)$$

$$(5) \text{ Symmetry: } x_{ij} = x_{ji}, \quad \forall i, j \in \{1, \dots, n\}, \quad (13)$$

$$(6) \text{ Diagonal: } x_{ii} = 0, \quad \forall i \in \{1, \dots, n\}. \quad (14)$$

1.2 Linear Program

$$x_{ij} \in \mathbb{R}_{\geq 0}, \quad \forall i, j \in \{1, \dots, n\}, \quad (15)$$

$$y_{ikj} \in \mathbb{R}_{\geq 0}, \quad \forall i, k, j \in \{1, \dots, n\}, \quad (16)$$

$$z_{ij} \in \mathbb{R}_{\geq 0}, \quad \forall i, j \in \{1, \dots, n\}, \quad (17)$$

Constraints:

$$(1) \text{ Indicator argument: } z_{ii} = \sum_{k=1}^n y_{iki} - \sum_{k=1}^n x_{ik}, \forall i \in \{1, \dots, n\}, \quad (18)$$

$$z_{ij} = \sum_{k=1}^n y_{ikj} + x_{ij}, \quad \forall i \neq j \in \{1, \dots, n\}, \quad (19)$$

$$(2) \text{ Indicator function: } z_{ij} \leq M g_{ij}, \quad \forall i, j \in \{1, \dots, n\}, \quad (20)$$

$$z_{ij} \geq g_{ij}, \quad \forall i, j \in \{1, \dots, n\}, \quad (21)$$

$$(3) \text{ Product: } y_{ikj} \leq x_{ik}, \quad \forall i, k, j \in \{1, \dots, n\}, \quad (22)$$

$$y_{ikj} \leq x_{kj}, \quad \forall i, k, j \in \{1, \dots, n\}, \quad (23)$$

$$y_{ikj} \geq x_{ik} + x_{kj} - 1, \quad \forall i, k, j \in \{1, \dots, n\} \quad (24)$$

$$(4) \text{ Upper Bound: } x_{ij} \leq 1, \quad \forall i, j \in \{1, \dots, n\}, \quad (25)$$

$$y_{ikj} \leq 1, \quad \forall i, k, j \in \{1, \dots, n\}, \quad (26)$$

$$(5) \text{ Symmetry: } x_{ij} = x_{ji}, \quad \forall i, j \in \{1, \dots, n\}, \quad (27)$$

$$(6) \text{ Diagonal: } x_{ii} = 0, \quad \forall i \in \{1, \dots, n\}. \quad (28)$$

1.3 Max Linear Program

$$x_{ij} \in \mathbb{R}_{\geq 0}, \quad \forall i, j \in \{1, \dots, n\}, \quad (29)$$

$$y_{ikj} \in \mathbb{R}_{\geq 0}, \quad \forall i, k, j \in \{1, \dots, n\}, \quad (30)$$

$$z_{ij} \in \mathbb{R}_{\geq 0}, \quad \forall i, j \in \{1, \dots, n\}, \quad (31)$$

Constraints:

$$(1) \text{ Indicator argument: } z_{ii} = \sum_{k=1}^n y_{iki} - \sum_{k=1}^n x_{ik}, \forall i \in \{1, \dots, n\}, \quad (32)$$

$$z_{ij} = \sum_{k=1}^n y_{ikj} + x_{ij}, \quad \forall i \neq j \in \{1, \dots, n\}, \quad (33)$$

$$(2) \text{ Indicator function: } z_{ij} \leq M g_{ij}, \quad \forall i, j \in \{1, \dots, n\}, \quad (34)$$

$$z_{ij} \geq g_{ij}, \quad \forall i, j \in \{1, \dots, n\}, \quad (35)$$

$$(3) \text{ Product: } y_{ikj} \leq x_{ik}, \quad \forall i, k, j \in \{1, \dots, n\}, \quad (36)$$

$$y_{ikj} \leq x_{kj}, \quad \forall i, k, j \in \{1, \dots, n\}, \quad (37)$$

$$y_{ikj} \geq x_{ik} + x_{kj} - 1, \quad \forall i, k, j \in \{1, \dots, n\} \quad (38)$$

$$(4) \text{ Upper Bound: } x_{ij} \leq 1, \quad \forall i, j \in \{1, \dots, n\}, \quad (39)$$

$$y_{ikj} \leq 1, \quad \forall i, k, j \in \{1, \dots, n\}, \quad (40)$$

$$(5) \text{ Symmetry: } x_{ij} = x_{ji}, \quad \forall i, j \in \{1, \dots, n\}, \quad (41)$$

$$(6) \text{ Diagonal: } x_{ii} = 0, \quad \forall i \in \{1, \dots, n\}, \quad (42)$$

Objective Function:

$$\max \sum_{i=1}^n \sum_{j=1}^n x_{ij}. \quad (43)$$

2 Integer Program Validity Proof

Define the edge set of H^2 as

$$E_{H^2} = \{uv \mid u \neq v, 1 \leq d_H(u, v) \leq 2\}. \quad (44)$$

Then for

$$A_G = \mathbb{1}_{\geq 1}((A_H)^2 - D_H + A_H). \quad (45)$$

Lemma 1. *The graph G defined by (45) is equivalent to the graph square H^2 defined in (44)*

$$ij \in E_{H^2} \iff 1 \leq d_H(i, j) \leq 2. \quad (46)$$

Proof. For $i \neq j$, since D_H is diagonal, have $(D_H)_{ij} = 0$ and hence

$$((A_H)^2 - D_H + A_H)_{ij} = (A_H)_{ij}^2 + (A_H)_{ij}.$$

Suppose $ij \in E_G$ such that $(A_H)_{ij}^2 + (A_H)_{ij} \geq 1$. This implies that at least one of $(A_H)_{ij} = 1$ or $(A_H)_{ij}^2 \geq 1$ must be true. If $(A_H)_{ij} = 1$, then $d_H(i, j) = 1$. If $(A_H)_{ij}^2 \geq 1$, then there exists a vertex k such that $ik, kj \in E_H$, and hence $d_H(i, j) \leq 2$. Therefore $1 \leq d_H(i, j) \leq 2$, and thus $ij \in E_{H^2}$.

Suppose $ij \in E_{H^2}$ such that $1 \leq d_H(i, j) \leq 2$. D_H is diagonal such that $(D_H)_{ij} = 0$. If $d_H(i, j) = 1$, then $(A_H)_{ij} = 1$. If $d_H(i, j) = 2$, then there exists a vertex k such that $ik, kj \in E_H$, and hence $(A_H)_{ij}^2 \geq 1$. In either case, $(A_H)_{ij}^2 + (A_H)_{ij} \geq 1$, so $(A_G)_{ij} = 1$, and hence $ij \in E_G$.

For $i = j$, since all graphs are simple, have $(E)_{ii} = 0$, $(A)_{ii} = 0$ and hence

$$((A_H)^2 - D_H + A_H)_{ii} = (A_H)_{ii}^2 - (D_H)_{ii}.$$

$(A_H)_{ii}^2 - (D_H)_{ii} = 0$ is always true since the degree matrix of H always has diagonal values $(D_H)_{ii} = \sum_k x_{ik}$ and each edge from vertex i to its neighbour in H will always create a loop to itself in the square such that $(A_H)_{ii}^2 = \sum_{k=1} x_{ik} x_{ki} = \sum_{k=1} x_{ik}^2$ by the symmetry of adjacency matrix A_H . Since $x_{ik} \in \{0, 1\}$, the square of the boolean is identical to the boolean x_{ik} such that $(A_H)_{ii}^2 = \sum_k x_{ik} = (D_H)_{ii}$ and $(A_G)_{ii} = \mathbb{1}_{\geq 1}((A_H)^2 - D_H + A_H)$ as required.

Combining both gives that $E_G = E_{H^2}$, and so $G = H^2$. $\square$

Theorem 1. *Let G be a graph and consider the integer program whose variables $x_{ij} \in \{0, 1\}$ encode a graph H with adjacency matrix $A_H = (x_{ij})$. Then G has a square root H if and only if the integer program has a feasible solution x such that the corresponding graph H satisfies $H^2 = G$.*

Proof. Suppose G has a square root H and let A_H be the adjacency matrix of H . $x_{ij} = (A_H)_{ij}$. Construct the variables x_{ij}, y_{ikj}, z_{ij} and the constant g_{ij} as such

- i) Let $x_{ij} = 1$ if $ij \in E_H$ and $= 0$ otherwise.
- ii) Let $y_{ikj} = x_{ik}x_{kj}$ for all i, k, j .
- iii) Let $z_{ij} = \sum_{k=1}^n y_{ikj} + x_{ij}$ for all i, j .
- iv) Let $z_{ii} = \sum_{k=1}^n y_{iki} - \sum_{k=1}^n x_{ik} = \sum_{k=1}^n x_{ik}^2 - \sum_{k=1}^n x_{ik} = 0$ since $x_{ij} \in \{0, 1\} \implies x_{ik}^2 = x_{ik}$.
- v) Let $g_{ij} = 1$ if $ij \in E_G$ and $= 0$ otherwise.

Verifying constraints of the integer program for the variables

- i) As $x_{ij} \in \{0, 1\} \implies y_{ikj} \in \{0, 1\}$, Equations (1)-(3) and Equations (11)-(12) are satisfied.
- ii) As H is a simple directed graph, its adjacency matrix is hallow and symmetric such that Equations (13) -(14) are satisfied.
- iii) As $y_{ikj} = x_{ik}x_{kj}$ and all variables are binary, Equations (8)- (10) are satisfied.
- iv) Equation (5) by definition is satisfied.
- v) As $z_{ii} = \sum_{k=1}^n x_{ik}x_{ki} - \sum_{k=1}^n x_{ik}$, Equation (4) is satisfied.
- vi) As $z_{ij} \geq 1 \iff x_{ij} = 1$ or $\exists k$ such that $x_{ik} = x_{kj} = 1$ and as $H^2 = G$, have $g_{ij} = 1 \iff 1 \leq d_H(i, j) \leq 2$. Thus $g_{ij} = 1 \iff z_{ij} \geq 1$ so the indicator constraints Equations (6)-(7) hold.

All constraints of the integer program are satisfied by the constructed variables. Hence, the integer program is feasible.

Suppose the integer program on graph G has a feasible solution x , and let x_{ij} define a graph H by

$$ij \in E_H \iff x_{ij} = 1.$$

As x_{ij} satisfies the constraints of the integer program,

- i) Equation (1) and Equation (11) gives $x_{ij} \in \{0, 1\}$ for all i, j ,
- ii) Equation (13) and Equation (14) give $x_{ij} = x_{ji}$ and $x_{ii} = 0$,

such that x_{ij} defines a hollow, symmetric matrix with all elements being either 0 or 1 which is a valid adjacency matrix of a graph. Examining the auxiliary variables y_{ikj} and z_{ij} ,

- i) Equations (8)-(10) give the strict equality $y_{ikj} = x_{ik}x_{kj}$ for all i, j, k ,
- ii) Equation (5) gives $z_{ij} = \sum_{k=1}^n y_{ikj} + x_{ij} = \sum_{k=1}^n x_{ik}x_{kj} + x_{ij}$ for all $i \neq j$,
- iii) Equation (4) gives $z_{ii} = \sum_{k=1}^n y_{iki} + \sum_{n} x_{ik} = \sum_{k=1}^n x_{ik}x_{ki} - \sum_{n} x_{ik} = 0$ for all i since

$$\sum_{k=1}^n x_{ik}x_{ki} = \sum_{k=1}^n x_{ik}^2 = \sum_{k=1}^n x_{ik},$$

by symmetry and since $x_{ij} \in \{0, 1\}$.

The indicator constraints Equations (6)-(7) thus give for $i \neq j$

$$g_{ij} = 1 \iff \sum_{k=1}^n x_{ik}x_{kj} + x_{ij} \geq 1 \text{ and } g_{ij} = 0 \iff \sum_{k=1}^n x_{ik}x_{kj} + x_{ij} = 0.$$

Since $x_{ij} \in \{0, 1\}$, this is equivalent to

$$g_{ij} = 1 \iff (x_{ij} = 1) \vee (\exists k \text{ such that } x_{ik} = 1 \wedge x_{kj} = 1) \text{ and } g_{ij} = 0 \iff x_{ij} = 0.$$

By definition of H this means that $g_{ij} = 1 \iff (ij \in E_H) \vee (\exists k \text{ such that } ik, kj \in E_H)$ which is equivalent to

$$g_{ij} = 1 \iff 1 \leq d_H(i, j) \leq 2.$$

Applying Lemma 1 thus gives $ij \in E_G \iff ij \in E_H^2$ such that $G = H^2$. $\square$

3 Two-Edge-Triangle Proof

Let x be an optimal solution to the linear relaxation defined by Equations (29)–(43). If $g_{ij} = 0$, then for every $k \in \{1, \dots, n\}$,

$$x_{ik} + x_{kj} = 1.$$

3.1 Integer Program

Lemma 2. *Consider the triangle with three vertices i, j and k where edges ik and kj are present and ij is absent in square G . Then for root G , the integer program defined by Equations (1)-(14) must satisfy*

$$x_{ik} + x_{kj} \leq 1.$$

Proof. As ij is absent in G , $g_{ij} = 0$ such that Equation (6) gives $z_{ij} \leq 0$ and Equation (7) gives $z_{ij} \geq 0$ such that $z_{ij} = 0$. Therefore Equation (5) gives

$$0 = \sum_{a=1}^n y_{iaj} + x_{ij} = \sum_{a=1}^n x_{ia}x_{aj} + x_{ij},$$

where the second equality follows from the product constraints Equations (8)–(10) together with the integrality of $x \in \{0, 1\}$ by Equation (1) and Equation (11). Since all variables are non-negative, 0 only holds when $x_{ij} = 0$ and all of $x_{ia}x_{aj} = 0$ and using the integrality of x , this means that at most one of x_{ia}, x_{aj} is equal to 1 and the other must be 0, which can be equivalently written as $x_{ia} + x_{aj} \leq 1$. Taking $a = k$ gives $x_{ik} + x_{kj} \leq 1$ as required. $\square$

3.2 Linear Program

Lemma 3. *Consider the triangle with three vertices i, j and k where edges ik and kj are present and ij is absent in square G . Then for root G , the linear program defined by Equations (15)-(28) must satisfy*

$$x_{ik} + x_{kj} \leq 1.$$

Proof. Since $g_{ij} = 0$, Equations (20)–(21) imply that $z_{ij} = 0$. Equation (19) therefore gives

$$0 = z_{ij} = \sum_{a=1}^n y_{iaj} + x_{ij}.$$

Since all variables are nonnegative by Equations (15)–(17), every term in the above sum must be equal to zero with $x_{ij} = 0$ and $y_{iaj} = 0$ for all a . Thus applying Equation (24) to $y_{ikj} = 0$ gives

$$0 = y_{ikj} \geq x_{ik} + x_{kj} - 1,$$

and hence $x_{ik} + x_{kj} \leq 1$ as required. $\square$

3.3 Max Linear Program

Proposition 1. *Consider the maximum linear relaxation defined by Equations (29)–(43) on the graph induced by only the three vertices $\{a, b, c\}$. Suppose*

$$g_{ab} = 1, \quad g_{ac} = 0, \quad g_{bc} = 1,$$

then there always exists an optimal, feasible solution satisfying $x_{ab} + x_{bc} = 1$.

Proof. Using Lemma 3, every feasible solution satisfies $x_{ab} + x_{bc} \leq 1$. Suppose to the contrary that $x_{ab} + x_{bc} = 1 - \delta$ where $\delta > 0$. As $x_{ac} = x_{ca} = 0$ by Lemma 3, Equation (41) gives $x_{ab} = x_{ba}$,

$x_{bc} = x_{cb}$ and Equation (42) gives $x_{aa} = x_{bb} = x_{cc} = 0$. Therefore the objective function can be simplified to

$$\sum_{u,v \in \{a,b,c\}} x_{uv} = x_{aa} + x_{ab} + x_{ac} + x_{ba} + x_{bb} + x_{bc} + x_{ca} + x_{cb} + x_{cc} = 2(x_{ab} + x_{bc}) = 2(1 - \delta).$$

Without loss of generality, increase x_{ab} such that $x'_{ab} = x_{ab} + \delta$ with the remaining $x'_{bc} = x_{bc}$ and $x'_{ac} = x_{ac} = 0$ remaining unchanged such that $x'_{ab} + x'_{bc} = x'_{ab} + x_{bc} = 1$. It remains to verify that there exist corresponding variables y' and z' satisfying Equations (30)–(35). Since only the variable x_{ab} has changed, using symmetry it is sufficient to examine every product variable whose upper or lower bounds involve x_{ab} .

A. Equation (32)

(i) Since $g_{aa} = 0$, Equations (34)–(35) imply $z'_{aa} = 0$. Therefore, by Equation (32)

$$0 = z'_{aa} = \sum_{k \in \{a,b,c\}} y'_{aka} - \sum_{k \in \{a,b,c\}} x'_{ak} = y'_{aaa} + y'_{aba} + y'_{aca} - x'_{aa} - x'_{ab} - x'_{ac}.$$

Since Equation (42) gives $x'_{aa} = 0$ and $x'_{ac} = 0$ by Lemma 3, Equations (36)–(37) imply $y'_{aaa} = 0$, $y'_{aca} = 0$. Therefore, $0 = y'_{aba} - x'_{ab} = y'_{aba} - (x_{ab} + \delta)$ and hence

$$y'_{aba} = x'_{ab} \tag{47}$$

is an additional constraint that is needed.

(ii) Since $g_{bb} = 0$, Equations (34)–(35) imply $z'_{bb} = 0$. Therefore, as in above

$$0 = z'_{bb} = \sum_{k \in \{a,b,c\}} y'_{bkb} - \sum_{k \in \{a,b,c\}} x'_{bk} = y'_{bab} + y'_{bcb} - x'_{ba} - x'_{bc}.$$

Since $x'_{ba} = x'_{ab}$ by Equation (41) and $x'_{bc} = x_{bc}$, it follows that

$$y'_{bab} + y'_{bcb} = x'_{ab} + x'_{bc} \tag{48}$$

is an additional constraint that is needed.

(iii) Since $g_{cc} = 0$, Equations (34)–(35) imply $z'_{cc} = 0$. Therefore, as in above

$$0 = \sum_{k \in \{a,b,c\}} y'_{ckc} - \sum_{k \in \{a,b,c\}} x'_{ck} = y'_{cac} + y'_{cbc} - x'_{ca} - x'_{cb}.$$

Since $x'_{ca} = 0 = x_{ca}$ by Lemma 3 and $x'_{cb} = x_{bc}$, this equation is unchanged and the original values of $y'_{cac} = y_{cac}$ and $y'_{cbc} = y_{cbc}$ remain feasible and no additional constraints are needed.

B. Equation (33)

(i) Since $g_{ab} = 1$, Equations (34)–(35) imply $z'_{ab} \geq 1$. Hence, by Equation (33),

$$1 \leq z'_{ab} = \sum_{k=1}^n y'_{akb} + x'_{ab} = y'_{aab} + y'_{abb} + y'_{acb} + x'_{ab} = y'_{acb} + x'_{ab},$$

since Equations (36)–(37) together with Equation (42) imply $y'_{aab} = y'_{abb} = 0$. As $1 \leq z_{ab}$, $z'_{ab} = z_{ab} + \delta$, $y'_{acb} = y_{acb}$ satisfies the new equation, no additional constraints are needed.

(ii) Since $g_{bc} = 1$, Equations (34)–(35) imply $z'_{bc} \geq 1$. Hence, by Equation (33),

$$1 \leq z'_{bc} = \sum_{k \in \{a,b,c\}} y'_{bk} + x'_{bc} = y'_{bac} + y'_{bbc} + y'_{bcc} + x'_{bc} = y'_{bac} + x'_{bc},$$

since Equations (36)–(37) together with Equation (42) imply $y'_{bbc} = y'_{bcc} = 0$. As $x'_{bc} = x_{bc}$, $y'_{bac} = y_{bac}$ satisfies the new equation, no additional constraints are needed.

(iii) Since $g_{ac} = 0$, Equations (34)–(35) imply $z'_{ac} = 0$. Hence, by Equation (33),

$$0 = z'_{ac} = \sum_{k \in \{a,b,c\}} y'_{ak} + x'_{ac} = y'_{aac} + y'_{abc} + y'_{acc} + x'_{ac} = y'_{abc},$$

since $y'_{aac} = y'_{acc} = 0$ by Equations (36)–(37), and $x'_{ac} = 0$ by Lemma 3. Therefore $y'_{abc} = y_{abc} = 0$ which is unchanged from the original feasible solution.

C. Equations (36)–(38). Consider any product variable y_{uvw} for which neither x_{uv} nor x_{vw} equals x_{ab} . Since these x -variables are unchanged, Equations (36)–(38) remain unchanged and are therefore still satisfied. Therefore only need to consider the product variables whose constraints involve x'_{ab}

$$y'_{aba}, y'_{abc}, y'_{cab}, y'_{bab}, y'_{acb}, y'_{bac}.$$

The remaining cases are analogous after interchanging a and b such that only three cases need to be examined $y'_{aba}, y'_{abc}, y'_{cab}$. However the previous equations give the constraints Equation (47) and Equation (48) hence also need to consider y'_{bab} and y'_{bcb} .

(i) Consider y'_{aba} . By Equation (47), $y'_{aba} = x'_{ab}$ which together with Equation (41) and Equation (39) giving $x'_{ab} \leq 1$ gives

$$y'_{aba} = x'_{ab} \leq x'_{ab}, \quad y'_{aba} = x'_{ab} \leq x'_{ba}, \quad y'_{aba} = x'_{ab} \geq 2x'_{ab} - 1 = x'_{ab} + x'_{ba} - 1.$$

Therefore Equations (36)–(38) are satisfied.

(ii) Consider y'_{bab} and y'_{bcb} . By Equation (48), $y'_{bab} + y'_{bcb} = x'_{ab} + x'_{bc}$ which is satisfied by the selection

$$y'_{bab} = x'_{ab}, \quad y'_{bcb} = x'_{bc}.$$

Equation (41) and Equation (39) giving $x'_{ab} \leq 1$, $x'_{bc} \leq 1$ gives

$$\begin{aligned} y'_{bab} = x'_{ab} &\leq x'_{ba}, & y'_{bab} = x'_{ab} &\leq x'_{ab}, & y'_{bab} = x'_{ab} &\geq x'_{ba} + x'_{ab} - 1 \\ y'_{bcb} = x'_{bc} &\leq x_{bc}, & y'_{bcb} = x'_{bc} &\leq x'_{cb}, & y'_{bcb} = x'_{bc} &\geq x'_{bc} + x'_{cb} - 1 \end{aligned}$$

Hence Equations (36)–(38) are satisfied.

(iii) Consider y'_{abc} . Since $x'_{ac} = 0$, Equations (36)–(37) imply that $y'_{abc} = 0$. As $x'_{ab} + x'_{bc} - 1 = 0$, so Equation (38) gives $y'_{abc} \geq 0$, which is unchanged from the original feasible solution.

(iv) Consider y'_{acb} . Since neither x'_{ac} nor x'_{cb} has been modified, all three product constraints are unchanged and therefore remain satisfied.

Therefore (x', y', z') can be constructed as

$$\begin{aligned} x'_{ab} &= x_{ab} + \delta, & x'_{ba} &= x_{ba} + \delta, & y'_{aba} &= x'_{ab}, & y'_{bab} &= x'_{ab}, & y'_{bcb} &= x_{bc}, \\ y'_{acb} &= y_{acb}, & y'_{bac} &= y_{bac}, & y'_{abc} &= 0, & z'_{ab} &= z_{ab} + \delta, & z'_{ba} &= z_{ba} + \delta, \end{aligned}$$

where every remaining variable is unchanged from (x, y, z) . As shown above, this is a feasible solution for Equations (29)–(35) and since

$$\sum_{u,v \in \{a,b,c\}} x'_{uv} = 2(x'_{ab} + x'_{bc}) = 2 > 2(1 - \delta) = \sum_{u,v \in \{a,b,c\}} x_{uv},$$

this contradicts the optimality of (x, y, z) and gives $x_{ab} + x_{bc} = 1$ as required. $\square$

Lemma 4. *Consider the maximum linear relaxation defined by Equations (29)–(43) on the graph induced by only the three vertices $\{a, b, c\}$ where $g_{ab} = g_{bc} = g_{ca} = 1$. Let (x, y, z) be an optimal feasible solution. If $x_{ab} < 1$, then there exists another optimal feasible solution (x', y', z') such that*

$$x'_{ab} = 1, \quad x'_{bc} = x_{bc}, \quad x'_{ca} = x_{ca}.$$

Consequently, within the induced triangle the value of each edge variable is independent of the values of the other two edge variables.

Proof. Suppose for contradiction $x_{ab} = 1 - \delta$ for some $\delta > 0$ and define $x'_{ab} = 1 > x_{ab}$ and $x'_{bc} = x_{bc}$, $x'_{ca} = x_{ca}$, it is necessary to verify that there exist corresponding variables y' and z' satisfying Equations (30)–(35) and that it is optimal with greater objective function than for (x, y, z) . By

A. Equation (32)

(i) Since $g_{aa} = 0$, Equations (34)–(35) imply $z'_{aa} = 0$. Therefore, by Equation (32)

$$0 = z'_{aa} = \sum_{k \in \{a,b,c\}} y'_{aka} - \sum_{k \in \{a,b,c\}} x'_{ak} = y'_{aaa} + y'_{aba} + y'_{aca} - x'_{aa} - x'_{ab} - x'_{ac}.$$

Since Equation (42) gives $x'_{aa} = 0$, Equations (36)–(37) imply $y'_{aaa} = 0$. Therefore, $0 = y'_{aba} - x'_{ab} = y'_{aba} - (x_{ab} + \delta)$ and hence

$$y'_{aba} + y'_{aca} = x'_{ab} + x'_{ac} \quad (49)$$

is an additional constraint that is needed.

(ii) Since $g_{bb} = 0$, Equations (34)–(35) imply $z'_{bb} = 0$. Therefore, as in above

$$0 = z'_{bb} = \sum_{k \in \{a,b,c\}} y'_{bkb} - \sum_{k \in \{a,b,c\}} x'_{bk} = y'_{bab} + y'_{bcb} - x'_{ba} - x'_{bc}.$$

Since $x'_{ba} = x'_{ab}$ by Equation (41) and $x'_{bc} = x_{bc}$, it follows that

$$y'_{bab} + y'_{bcb} = x'_{ab} + x'_{bc} \quad (50)$$

is an additional constraint that is needed.

(iii) Since $g_{cc} = 0$, Equations (34)–(35) imply $z'_{cc} = 0$. Therefore, as in above

$$0 = \sum_{k \in \{a,b,c\}} y'_{ckc} - \sum_{k \in \{a,b,c\}} x'_{ck} = y'_{cac} + y'_{cbc} - x'_{ca} - x'_{cb}.$$

Since $x'_{cb} = x_{bc}$ and $x'_{ca} = x_{ca}$ by Equation (41), it follows that

$$y'_{cac} + y'_{cbc} = x'_{ac} + x'_{bc} \quad (51)$$

is an additional constraint that is needed.

B. Equation (33)

(i) Since $g_{ab} = 1$, Equations (34)–(35) imply $z'_{ab} \geq 1$. Hence, by Equation (33),

$$1 \leq z'_{ab} = \sum_{k=1}^n y'_{akb} + x'_{ab} = y'_{aab} + y'_{abb} + y'_{acb} + x'_{ab} = y'_{acb} + x'_{ab},$$

since Equations (36)–(37) together with Equation (42) imply $y'_{aab} = y'_{abb} = 0$. As $1 \leq z_{ab}$, $z'_{ab} = z_{ab} + \delta$, $y'_{acb} = y_{acb}$ satisfies the new equation, no additional constraints are needed.

(ii) Since $g_{bc} = 1$, Equations (34)–(35) imply $z'_{bc} \geq 1$. Hence, by Equation (33),

$$1 \leq z'_{bc} = \sum_{k \in \{a,b,c\}} y'_{bkc} + x'_{bc} = y'_{bac} + y'_{bbc} + y'_{bcc} + x'_{bc} = y'_{bac} + x'_{bc},$$

since Equations (36)–(37) together with Equation (42) imply $y'_{bbc} = y'_{bcc} = 0$. As $x'_{bc} = x_{bc}$, $y'_{bac} = y_{bac}$ satisfies the new equation, no additional constraints are needed.

(iii) Since $g_{ac} = 1$, Equations (34)–(35) imply $z'_{ac} \geq 1$. Hence, by Equation (33),

$$1 \leq z'_{ac} = \sum_{k \in \{a,b,c\}} y'_{akc} + x'_{ac} = y'_{aac} + y'_{abc} + y'_{acc} + x'_{ac} = y'_{abc} + x'_{ac},$$

since Equations (36)–(37) together with Equation (42) imply $y'_{aac} = y'_{acc} = 0$. As $x'_{ac} = x_{ac}$, $y'_{abc} = y_{abc}$ satisfies the new equation, no additional constraints are needed.

C. Equations (49)–(51) can satisfied with the feasible solution

$$y'_{aba} = x'_{ab} = 1, \quad y'_{aca} = x'_{ac}, y'_{bab} = x'_{ab} = 1, \quad y'_{bcb} = x'_{bc}, \quad y'_{cac} = x'_{ac}, \quad y'_{cbc} = x'_{bc}.$$

Thus it only remains to verify that Equations (36)–(38) hold. Consider any product variable y_{uvw} for which neither x_{uv} nor x_{vw} equals x_{ab} . Since these x -variables are unchanged, Equations (36)–(38) remain unchanged and are therefore still satisfied. Therefore only need to consider the product variables whose constraints involve x'_{ab}

$$y'_{aba}, y'_{abc}, y'_{cab}, y'_{bab}, y'_{acb}, y'_{bac}.$$

Only three cases need to be examined $y'_{aba}, y'_{abc}, y'_{cab}$ which must satisfy must also be satisfied.

(i) Consider y'_{aba} and y'_{aca} . By Equation (49), $y'_{aba} + y'_{aca} = x'_{ab} + x'_{ac}$ which is satisfied by the selection

$$y'_{aba} = x'_{ab}, \quad y'_{aca} = x'_{ac}.$$

Equation (41) and Equation (39) giving $x'_{ab} \leq 1$, $x'_{ac} \leq 1$ gives

$$y'_{aba} = x'_{ab} \leq x'_{ba}, \quad y'_{aba} = x'_{ab} \leq x'_{ab}, \quad y'_{aba} = x'_{ab} \geq x'_{ab} + x'_{ba} - 1$$

$$y'_{aca} = x'_{ac} \leq x_{ac}, \quad y'_{aca} = x'_{ac} \leq x'_{ac}, \quad y'_{aca} = x'_{ac} \geq x'_{ac} + x'_{ca} - 1$$

Hence Equations (36)–(38) are satisfied.

- (ii) Consider y'_{bab} and y'_{bcb} . By Equation (48), $y'_{bab} + y'_{bcb} = x'_{ab} + x'_{bc}$ which is satisfied by the selection

$$y'_{bab} = x'_{ab}, \quad y'_{bcb} = x'_{bc}.$$

Equation (41) and Equation (39) giving $x'_{ab} \leq 1$, $x'_{bc} \leq 1$ gives

$$y'_{bab} = x'_{ab} \leq x'_{ba}, \quad y'_{bab} = x'_{ab} \leq x'_{ab}, \quad y'_{bab} = x'_{ab} \geq x'_{ba} + x'_{ab} - 1$$

$$y'_{bcb} = x'_{bc} \leq x_{bc}, \quad y'_{bcb} = x'_{bc} \leq x'_{cb}, \quad y'_{bcb} = x'_{bc} \geq x'_{bc} + x'_{cb} - 1$$

Hence Equations (36)–(38) are satisfied.

- (iii) Consider y'_{abc} . Since $x'_{ab} = 1$, Equations (36)–(38) give

$$y'_{abc} \geq x'_{ab} + x'_{bc} - 1 = x'_{bc}, \quad y'_{abc} \leq x'_{ab}, \quad y'_{abc} \leq x'_{bc}.$$

If $x'_{bc} < 1$, then $y'_{abc} = x'_{bc}$ satisfies all constraints and if $x'_{bc} = 1$ then $y'_{abc} = 1$ satisfies all constraints.

- (iv) Consider y'_{acb} . Since neither x'_{ac} nor x'_{cb} has been modified, all three product constraints are unchanged and therefore remain satisfied.

- D. Equations (36)–(38). Only the product variables involving the modified edge variable x'_{ab} require verification. These are

$$y'_{aba}, y'_{abc}, y'_{cab}, y'_{bab}, y'_{acb}, y'_{bac}.$$

The remaining product variables satisfy exactly the same constraints as before and therefore remain feasible.

The equations obtained in Parts A(i)–A(iii), (49), (50), and (51), must also be satisfied. A feasible choice is

$$\begin{aligned} y'_{aba} &= 1, & y'_{aca} &= x'_{ac}, \\ y'_{bab} &= 1, & y'_{bcb} &= x'_{bc}, \\ y'_{cac} &= x'_{ac}, & y'_{cbc} &= x'_{bc}. \end{aligned}$$

These satisfy the three linear equations immediately.

It therefore remains only to verify $y'_{abc}, y'_{cab}, y'_{acb}$ and y'_{bac} .

For y'_{abc} , since $x'_{ab} = 1$, the product constraints reduce to

$$x'_{bc} \leq y'_{abc} \leq x'_{bc},$$

so necessarily

$$y'_{abc} = x'_{bc}.$$

Similarly,

$$y'_{bac} = x'_{bc}, \quad y'_{cab} = x'_{ac}, \quad y'_{acb} = x'_{ac}.$$

Hence every product constraint is satisfied.

Therefore (x', y', z') can be constructed as

$$\begin{aligned} x'_{ab} &= 1, & x'_{ba} &= 1, & y'_{aba} &= 1, & y'_{aca} &= x'_{ac}, \\ y'_{bab} &= 1, & y'_{bcb} &= x'_{bc}, & z'_{ab} &= z_{ab} + \delta, & z'_{ba} &= z_{ba} + \delta, \end{aligned}$$

where every remaining variable is unchanged from (x, y, z) . As shown above, this is a feasible solution for Equations (29)–(35) and since

$$\sum_{u,v \in \{a,b,c\}} x'_{uv} = 2(x'_{ab} + x'_{bc} + x'_{ac}) = 2(x_{ab} + \delta + x'_{bc} + x'_{ac}) > 2(x_{ab} + x'_{bc} + x'_{ac}) = \sum_{u,v \in \{a,b,c\}} x_{uv},$$

this contradicts the optimality of (x, y, z) and gives $x_{ab} = 1$ as required. Since the choice of edge ab was arbitrary, the same argument applies to the remaining two edges. Therefore every edge variable of the induced triangle attains the value 1 in an optimal solution, while the value of each edge can be increased independently of the other two. $\square$

Proposition 2. *Consider the maximum linear relaxation defined by Equations (29)–(43) on the square graph G . Suppose*

$$g_{ik} = 1, \quad g_{kj} = 1, \quad g_{ij} = 0,$$

and let (x, y, z) be a feasible solution satisfying $x_{ik} + x_{kj} < 1$. Then in order for there to exist another feasible solution (x', y', z') such that $\sum_{u,v \in \{i,j,k\}} x'_{uv} > \sum_{u,v \in \{i,j,k\}} x_{uv}$, G must be infinitely large.

Proof. Using Proposition 1, any three vertices that satisfy the maximum linear relaxation defined by Equations (29)–(43) will always satisfy $x_{ik} + x_{kj} = 1$. Hence if there exists three vertices i, j, k with a feasible solution satisfying the relationship $x_{ik} + x_{kj} = 1 - \delta$ where $\delta > 0$, then increasing either x_{ik} or x_{kj} must violate a constraint involving a vertex $\ell \notin \{i, j, k\}$.

- A. If ℓ is connected to $\{i, j, k\}$ by only one vertex i , then can always increase x_{kj} and vice versa hence can only do so via k .
- B. If ℓ is connected to $\{i, j, k\}$ by two vertices, then if one of i or j is chosen together with k a triangle will be formed

$\square$

Theorem 2. *Let x be an optimal solution to the linear relaxation defined by Equations (29)–(43). If $g_{ij} = 0$, then for every $k \in \{1, \dots, n\}$,*

$$x_{ik} + x_{kj} = 1.$$

Proof. Using Lemma 3 $x_{ik} + x_{kj} \leq 1$ and using Proposition 2 $x_{ik} + x_{kj} \geq 1$ such that $x_{ik} + x_{kj} = 1$ as required. $\square$